

UNIT-1: WATER CHEMISTRY

INTRODUCTION:

Water, the Earth's most plentiful resource, is indispensable for the existence of all living organisms, including humans, animals, and plants. Beyond its critical role in sustaining life, water serves as a versatile utility in numerous industries, functioning as a coolant, solvent, generator of steam, and a vital component in processes like air conditioning and fire suppression. Furthermore, water is a cornerstone of our daily lives, serving various domestic needs such as bathing, drinking, washing, sanitation, and irrigation.

Types of Impurities present in the water or characteristics imparted by impurities in water

The natural water is usually contaminated by different types of impurities.

They are mainly three types:

- Physical impurities
- Chemical impurities
- Biological impurities

1. Physical impurities:

- ❖ **Colour:** The colour in water is caused by metallic substances like salts of iron, manganese, industrial effluents. This is due to dissolved substances and substances present as fine colloids. Example: yellowish-red colour indicates the presence of Iron.
- ❖ **Turbidity:** is due to the colloidal, extremely fine suspensions such as insoluble substances like clay, silt, and micro-organisms. Turbidity in water can be eliminated by sedimentation, followed by coagulation, filtration, etc.
- ❖ **Taste:** The presence of dissolved minerals in water produces taste. Bitter taste can be due to the presence of Fe, Al, Mn, Sulphates and lime. Soap taste can be due to the presence of large amount of sodium bicarbonate.
- ❖ **Odour:** In water is undesirable for domestic as well as industrial purpose. Disagreeable odour in water may be caused by the presence of living organisms, decaying vegetation including algae, bacteria and fungi.

2. Chemical impurities:

It includes: *Inorganic and organic chemicals released from dyes, paints, vanishes, drugs, insecticides, pesticides, etc. All these pollute water bodies.*

- ❖ Acids discharged in water by DDT, high explosives, battery, industries, etc. The use of this type of contaminated water cause harmful effects on health of human-beings.

3. Biological Impurities:

- Biological impurities are algae, pathogenic bacteria, fungi, viruses, pathogens, parasite-worms, etc. The sources of these contamination is discharge of domestic and sewage wastes, excreta (from man, animals and birds), etc.

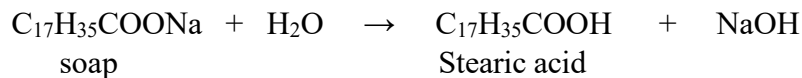
HARDNESS OF WATER

Hardness of water is defined as which prevents the lathering of soap. This is due to presence of certain

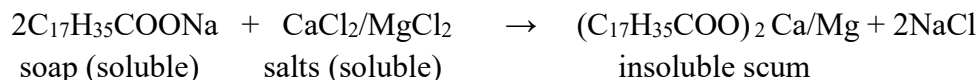
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salts like Ca^{+2} , Mg^{+2} and other heavy metals dissolved in water. Soaps (Sodium or Potassium salts of higher fatty acids) like Stearic acids ($\text{C}_{17}\text{H}_{35}\text{COONa}$).

Soft Water: The water which gives more lather with soap is called soft water.



Hard Water: The water which does not give lather with soap is called hard water. This is due to presence of certain salts like Ca^{+2} , Mg^{+2} and other heavy metals dissolved in water

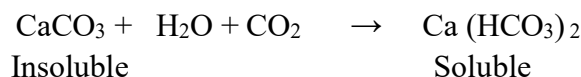


CAUSES OF HARDNESS

Hardness of water is due to the presence of Bicarbonates, Chlorides, Sulphates and Nitrates of Calcium and Magnesium. These soluble salts get mixed with natural water due to the following reasons:

1. When natural water containing CO_2 flows over the rocks of the limestone (CaCO_3) and Dolomite

(CaCO_3 & MgCO_3), they get converted into soluble bicarbonates. Thus, water gets hardness.



2. When natural water flows over the rocks containing chlorides and sulphates and Nitrates of Calcium and magnesium, these salts dissolve in water. Thus water gets hardness.

DISADVANTAGES OF HARDNESS

1. In Domestic use:

- ❖ **Washing:** Hard water, when used for washing purposes, does not producing lather freely with soap. As a result cleaning quality of soap is decreased and a lot of it is wasted.
- ❖ **Bathing:** Hard water does not lather freely with soap solution, but produces sticky scum on the bath-tub and body. Thus, the cleaning quality of soap is depressed and a lot of it is wasted.
- ❖ **Cooking:** The boiling point of water is increased because of presence of salts. Hence more fuel and time are required for cooking.
- ❖ **Drinking:** Hard water causes bad effects on our digestive system. Moreover, the possibility of forming calcium oxalate crystals in urinary tracks is increased.

2. Industrial Use:

- ❖ **Textile Industry:** Hard water causes wastage of soap. Precipitates of calcium and magnesium soaps adhere to the fabrics and cause problem.

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- ❖ **Sugar Industry:** The water which containing sulphates, nitrates, alkali carbonates are used in sugar refining, cause difficulties in the crystallization of sugar.
- ❖ **Dyeing Industry:** The dissolved salts in hard water may reacts with costly dyes forming precipitates.
- ❖ **Paper Industry:** Calcium, magnesium, Iron salts in water may affect the quality of paper.
- ❖ **Pharmaceutical Industry:** Hard water may cause some undesirable products while preparation of pharmaceutical products.

3. Steam generation in Boilers:

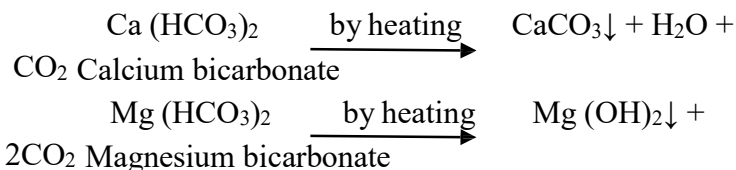
- ❖ For steam generation, boilers are almost invariably employed. If the hard water is fed directly to the boilers, there arise many troubles such as: Scales & sludges formation, Corrosion, Priming & Foaming and Caustic embrittlement.

TYPES OF HARDNESS

Hardness of water is mainly two types:

1. Temporary Hardness
2. Permanent Hardness

1. Temporary or Carbonate Hardness: Temporary Hardness mainly caused by the presence of dissolved bicarbonates of Calcium, Magnesium ($\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$). Temporary Hardness can be largely removed by boiling of water.



2. Permanent or Non-Carbonate Hardness: It is due to the presence of dissolved Chlorides, Nitrates and Sulphates of Calcium, Magnesium, Iron and other metals. Permanent hardness responsible salts are CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4 , FeSO_4 , $\text{Al}_2(\text{SO}_4)_3$. Permanent Hardness cannot be removed by boiling but it can be removed by the use of *chemical agents*.

EQUIVALENTS OF CALCIUM CARBONATE

The expression of hardness producing salts usually expressed in terms of an equivalent amount of CaCO_3 . Calcium Carbonate is chosen as a standard because:

- i. Its molecular weight (100) and equivalent weight (50) is a whole number, so the calculations in water analysis can be simplified.
- ii. It is the most insoluble salt that can be precipitated in water treatment.

The conversion of the hardness causing salts into CaCO_3 equivalents can be achieved by using the following formula:

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The equivalent of $\text{CaCO}_3 = \frac{\text{The weight of hardness causing salts}}{\text{Molecular weight of hardness causing salts}} \times 100$ (Molecular weight of CaCO_3)

Table: Calculations of equivalents of Calcium Carbonate (CaCO_3)

<i>Dissolved salt/ion</i>	<i>Molar mass</i>	<i>Chemical equivalent</i>	<i>Multiplication factor for converting into equivalents of CaCO_3</i>
$\text{Ca}(\text{HCO}_3)_2$	162	81	100/162
$\text{Mg}(\text{HCO}_3)_2$	146	73	100/146
CaSO_4	136	68	100/136
CaCl_2	111	55.5	100/111
MgSO_4	120	60	100/120
MgCl_2	95	47.5	100/95
CaCO_3	100	50	100/100
MgCO_3	84	42	100/84
CO_2	44	22	100/44
$\text{Ca}(\text{NO}_3)_2$	164	82	100/164
$\text{Mg}(\text{NO}_3)_2$	148	74	100/148
HCO_3^-	61	61	100/122
OH^-	17	17	100/34
CO_3^{2-}	60	30	100/60
NaAlO_2	82	82	100/164
$\text{Al}_2(\text{SO}_4)_3$	342	57	100/114
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	278	139	100/278
H^+	1	1	100/2
HCl	36.5	1	100/73

UNITS OF HARDNESS

1. Parts per Million (ppm): The number of parts of calcium carbonate equivalent hardness presents in 10^6 parts of water.

$$1\text{ ppm} = 1 \text{ part of CaCO}_3 \text{ eq hardness in } 10^6 \text{ parts of water.}$$

2. Milligrams per litre (mg/l): The number of milligrams of calcium carbonate equivalent hardness presents in litre of water.

$$1 \text{ mg/L} = 1 \text{ mg of CaCO}_3 \text{ eq hardness in 1 litre of water.}$$

But one litre of water weights = 1 kg = 1000g = 1000 x 1000 mg = 10^6 mg = 1 ppm.

3. Clark's degree ($^{\circ}\text{Cl}$): The number of parts of calcium carbonate equivalent hardness presents in 70,000 or (7×10^4) parts of water.

$$1^{\circ} \text{ Clarke} = 1 \text{ part of CaCO}_3 \text{ eq hardness per 70,000 parts of water.}$$

4. Degree French ($^{\circ}\text{Fr}$): The number of parts of calcium carbonate equivalent hardness presents in 10^5 parts of water.

$$1^{\circ} \text{ Fr} = 1 \text{ part of CaCO}_3 \text{ hardness eq per } 10^5 \text{ parts of water.}$$

Relationship between various units of hardness:

1 ppm	= 1 mg/L	= 0.1° Fr	= 0.07° Cl
1 mg/L	= 1 ppm	= 0.1° Fr	= 0.07° Cl
1° Cl	= 1.433° Fr	= 14.3 ppm	= 14.3 mg/L
1° Fr	= 10 ppm	= 10 mg/L	= 0.7° Cl

BOILER TROUBLES/PROBLEMS

A boiler is a closed vessel in which water under pressure is transformed into steam by the application of heat. The steam so generated is used in industries and generation of power. In modern pressure boilers and laboratories, the water required is used pure than the distilled water.

A boiler feed water should correspond with the following composition:

- Its hardness should be below 0.2ppm.
- Its caustic alkalinity (due to OH^-) should lie between 0.15ppm to 0.45ppm.
- It's should be free from dissolved gases like O_2 , CO_2 , in order to prevent boiler corrosion.

Excess of impurities in the boiler feed water generally cause the following problems:

- 1. Sludge's and Scale formation**
- 2. Priming and Foaming**
- 3. Caustic embrittlement**
- 4. Boiler Corrosion**

Boilers are employed for the steam generation in power plants, where water is continuously heated to produce steam. As more and more water is removed from water in the form of steam, the boiler water gets concentrated with dissolved salts progressively reaches the saturation point. At this point the dissolved salts are precipitated out and slowly settle on the inner walls of the boiler plate. The precipitation takes place in two ways.

1. SLUDGES AND SCALES FORMATION

SLUDGES:

Sludge is a soft, loosy and slimy precipitate formed within the boiler. It is formed at comparatively colder portions of the boiler and collects in the area where flow rate is slow.

Ex: MgCO_3 , MgCl_2 , CaCl_2 , MgSO_4 .

Reasons for formation of sludges:

The dissolved salts whose solubility is more in hot water and less in cold water produce sludges.

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Disadvantages of sludges:

1. Sludges are bad conductors of heat and results in the wastage of heat and fuel.
2. Excessive sludge formation leads to the settling of sludge in slow circulation areas such as pipe connections, plug openings, gauge-glass connections leading to the choking of the pipes.

Prevention of sludge formation:

- a. By using soft water which is free from dissolved salts like MgCO_3 , MgCl_2 , CaCl_2 and MgSO_4 can be prevent sludge formation.
- b. By blow down operation carried out frequently can prevent sludge formation.

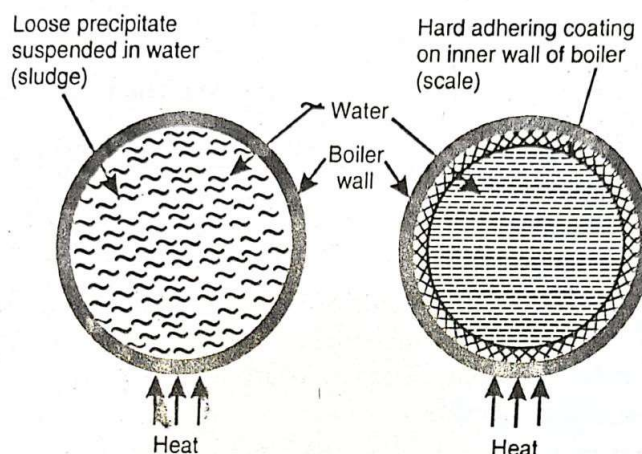


Fig. Scale and Sludge in boilers

Scales are hard, adhering precipitates formed on the inner walls of the boilers. Scales are stick very firmly on to the inner walls of the boiler. It is removed with chisel and hammer. Scales are formed by decomposition of calcium bicarbonate in low pressure boilers.

SCALES:

Reasons for formation of scales:

- a. **Decomposition of calcium bicarbonate:** The calcium bicarbonate at high temperature decomposes to calcium carbonate which is insoluble salt, forms scale in low pressure boilers.



- b. **Hydrolysis of Magnesium salts:** Magnesium salts gets hydrolyzed at high temperature forming $\text{Mg}(\text{OH})_2$ precipitation which forms salt type scale.



- c. **Decomposition of calcium sulphate:** The solubility of CaSO_4 in water decreases with the increase in temperature and forms precipitation on the surface of the boiler further which forms hard scale. This

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type of scales is formed in high-pressure boilers.

- d. **Presence of silica:** SiO_2 present even in small quantities, deposits as Calcium silicates (CaSiO_3) or Magnesium silicates (MgSiO_3). The deposits form hard scale and are very difficult to remove.

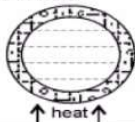
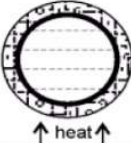
Disadvantages of Scales:

1. **Wastage of heat and fuels:** Scales poor thermal conductivity so that rate of heat transformation is reduced.
2. **Lowering of boiler safety** is due to overheating of the boiler material becomes softer and weaker, which causes distortion of boiler.
3. **Decrease in efficiency** of the boiler due to scales deposited in the valves and condensers of the boiler cause choking.
4. **Danger of explosion** which happens the formation of the scales, the boiler plate faces higher temperature outside and lesser temperature inside due to uneven expansion. The water comes suddenly contact with overheated portion and larger amount steam is formed immediately, this results in development of sudden high pressure which may cause explosion of the boiler.

Prevention of scales:

- a. If the scale formation is soft it can be removed by a scrapper, wire brush.
- b. By giving thermal shocks, by sudden heating and sudden cooling which makes scale brittle and removed by scrubbing with wire brush.
- c. If scale is very hard that is formed by CaCO_3 can be removed by washing with 5-10% HCl and CaSO_4 can be removed with EDTA solution.

Difference between Sludge and Scale:

No	Sludge	Scale
1	Loose, slim , non-adherent precipitate	Hard, thick , strong adherent precipitate
2	Due to salts like MgSO_4 , MgCl_2	Due to salts like CaSO_4 , $\text{Ca}(\text{HCO}_3)_2$
3	Due to poor conductance, they decrease the boiler efficiency to lesser extent and causing choking in the pipelines.	Due to poor conductance, they decrease the boiler efficiency to maximum extent, cause reduced fuel economy , improper boiling, boiler explosion etc.,
4	It can be prevented by periodical replacement of concentrated hard water by fresh water. This process is known as "blow down" method.	It can be prevented by special methods like i)external treatment of ion exchange , ii)Internal carbonate, phosphate, Calgon conditioning iii)Mechanical hard scrubbing methods.
5	Diagram: Loose Precipitate Suspended in water (sludge) 	 Adherent coating inside the boiler (scale)

2. PRIMING AND FOAMING

PRIMING:

When a boiler is steaming (i.e., producing steam) rapidly, some particles of the liquid water are carried along with the steam. This process of '*wet steam*' formation is known as '**Priming**'.

Causes:

Priming is caused by:

1. The presence of large amount of dissolved salts
2. High steam velocities
3. Sudden boiling
4. Improper boiler design
5. Sudden increase in steam production rate.

Prevention:

1. By fitting mechanical purifiers
2. Avoiding rapid change in steam rate
3. Maintaining low water levels in boilers
4. Effective softening and filtration of boiler feed water
5. Blow down of the boiler.

FOAMING:

The phenomenon of formation of *persistent foam or bubbles* on the surface of water inside the boiler, which do not break easily.

Causes:

1. Pure water has very little foaming whereas the water containing dissolved impurities and suspended matter has a greater tendency to produce foam.
2. The presence of large quantity of suspended impurities and oils lowers the surface tension producing foam.

Prevention:

1. Adding antifoaming chemicals like castor oil.
2. Besides castor oil, Gallic acid and tannic acids, corn oil, cotton seed oil, sperm oil, bees wax, etc., are also used as antifoaming agents.
3. Blow down operation of the boiler can prevent foaming.
4. Removing oil from boiler water by adding compounds like sodium aluminate.

Disadvantages of Priming and Foaming:

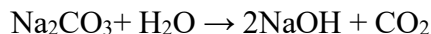
Priming and Foaming usually occur together. They are objectionable because: -

1. The dissolved salts in boiler water are carried by the wet steam to super heater and turbine blades where they get deposited and reduce the efficiency.
2. The dissolved salts enter the parts of other machinery where steam is used thereby decreasing the life of machinery.
3. The actual height of the water column cannot be judged thereby making the maintenance of boiler pressure becomes difficult.

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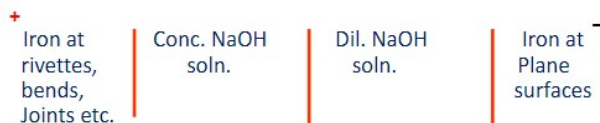
3. CAUSTIC EMBRITTLEMENT

The formation of brittle and in crystalline cracks in the boiler shell is called caustic embrittlement. The main reason for this is the presence of alkali-metal carbonates and bicarbonates in feed water. In lime-soda process, it is likely that, some residual Na_2CO_3 is still present in the softened water. This Na_2CO_3 decomposes to give NaOH and CO_2 , due to which the boiler water becomes “Caustic Soda”.



The H₂O evaporates, the concentration of NaOH increase progressively creating a concentration cell as given below thus dissolving the iron of the boiler as sodium ferrate (Na₂FeO₂).

Caustic attack on boiler parts can be represented as:



This causes embrittlement of boiler parts such as bends, joints, reverts etc, due to which the boiler gets fail. The iron at plane surfaces surrounded by dilute NaOH becomes cathodic while the iron at bends and joints surrounded by highly concentrated NaOH becomes anodic which consequently decayed or corroded.

Caustic embrittlement can be prevented:

- a. By maintaining the pH value of water and neutralization of alkali.
- b. By using Sodium Phosphate as softening reagents, in the external treatment of boilers.
- c. Caustic embrittlement can also be prevented by adding Tannin or Lignin or Sodium sulphate which prevents the infiltration of caustic-soda solution blocking the hair-cracks.

4. BOILER CORROSION

Boiler corrosion:

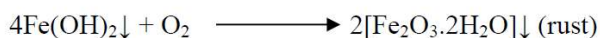
The decay of boiler material by chemical or electrochemical attack by its environment is known as “**Boiler corrosion**”.

The main reasons for boiler corrosion are:

1. **Dissolved Oxygen:** water contains usually about 8ml of dissolved oxygen per liter at room temperature.

Disadvantages of dissolved oxygen:

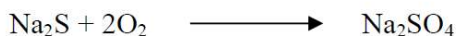
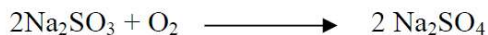
In presence of prevailing high temperature, the dissolved Oxygen attacks boiler material as follows:



Removal of dissolved Oxygen:

1) By adding calculated quantity of Sodium sulphite (Na_2SO_3) or hydrazine (N_2H_4) or sodium sulphide (Na_2S).

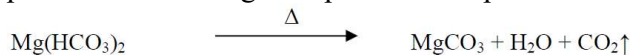
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Hydrazine is an ideal internal treatment chemical for the removal of dissolved oxygen.

Azamina 8001-RD (a polyvalent organic compound) has been employed for degassing water in minimum time.

2. Dissolved Carbon dioxide: has a slow corrosive effect on the materials of boiler plate. Under the high temperature and pressure the bicarbonates decompose to produce CO_2 .



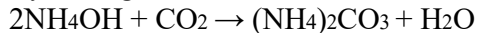
Disadvantages of CO_2 :

It has slow corrosive effect on boiler plates by producing the carbonic acid.



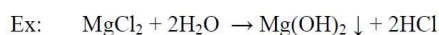
Removal of CO_2 :

By adding calculated amount of ammonia.



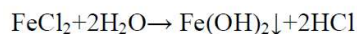
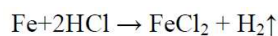
By mechanical deaeration process along with oxygen.

3. Acids from dissolved salts: water containing dissolved Mg salts liberate acids on hydrolysis.



Disadvantages of acids:

The acids react with the iron of boiler plate in a chain reaction producing HCl again and again.



Presence of even a small amount of MgCl_2 will have corrosion of iron to a large extent.

Prevention of acid corrosion:

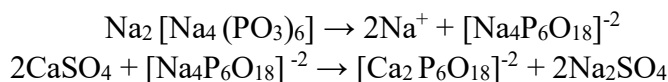
- 1) By softening of boiler water to remove MgCl_2 from the water.
 - 2) By frequent blow down operation.
 - 3) Addition of inhibitors to form a thin film on the surface of the boiler protecting the metal from attack.
- Sodium silicates, sodium phosphates and sodium chromate acts as good corrosion inhibitors.

INTERNAL TREATMENT OF WATER

Suitable chemicals are added to the boiler water either to precipitate or to convert the scale into compounds is called **internal treatment** of the boiler feed water. Internal treatment can be done following types.

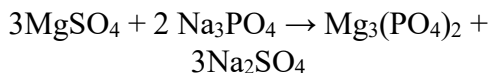
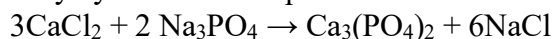
1. Calgon conditioning: Involves in adding calgon to boiler water. It prevents the scale and sludge formation by forming soluble complex compound with CaSO_4 .

Calgon = Sodium hexa meta phosphate = $\text{Na}_2 [\text{Na}_4 (\text{PO}_3)_6]$



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2. Phosphate conditioning: The addition of sodium phosphate in hard water reacts with the hardness causing agents and gives calcium and magnesium phosphates which are soft and non-adhere and can be removed easily by blow-down operation.

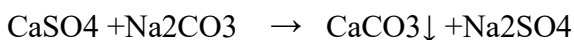


Generally three types of Phosphates are employed.

- Tri sodium Phosphate (Na_3PO_4): is too alkaline used for treat to too acidic water.
- Di sodium Phosphate ($\text{Na}_2 \text{HPO}_4$): is weakly alkaline used for treat to weakly acidic water.
- Sodium dihydrogen Phosphate ($\text{Na H}_2\text{PO}_4$): is too acidic used for treat to too alkaline water.

3. Colloidal conditioning: The addition of organic substances such as Kerosene, tannin, Gel. These substances gets coated over the scale forming precipitates and gives a loose and non-sticky precipitates which can be removed by using blow-down operation.

4. Carbonate conditioning: Reagent added is sodium carbonate. In low pressure boilers scale formation can be avoided by adding sodium carbonate to boiler water.



Sludge

Deposition of CaSO_4 as scale doesn't takes place and calcium is precipitated as loose sludge of CaCO_3 which can be removed by blow down operation.

EXTERNAL TREATMENT:

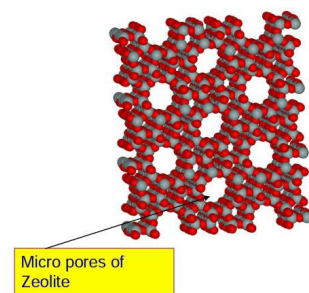
Water used for industrial purposes (such as steam generation) should be sufficiently pure. it should, therefore, be free from hardness-producing salts before it is put to use. Removing hardness-producing salts from water is known as softening of water. In industry, four methods are mainly employed for softening of water.

1. ZEOLITE OR PERMUTIT PROCESS

Zeolite are hydrated sodium alumino silicates $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$. ($x=2-10$ and $y=2-6$) (inorganic salts). They work as water softners by replacing the calcium and magnesium ions in water with the sodium ions in zeolite.

1. Natural zeolites Synthetic zeolites Non-porous, amorphous and durable e.g. $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{SiO}_2 \cdot 2\text{H}_2\text{O}$
 $\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot 4\text{H}_2\text{O}$

2. Synthetic zeolites (Natrolite) (Laumonite) Gel like structures, generally porous Na_2CO_3 , Al_2O_3 and SiO_2



Porous Structure of zeolite

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heated together $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$. ($x=2-10$ and $y=2-6$)

In this process hard water is allowed to pass through a bed of zeolite at a specified rate. Then the sodium ions present in the zeolite bed continuously replace the calcium and magnesium ions present in water and hence the water becomes soft.

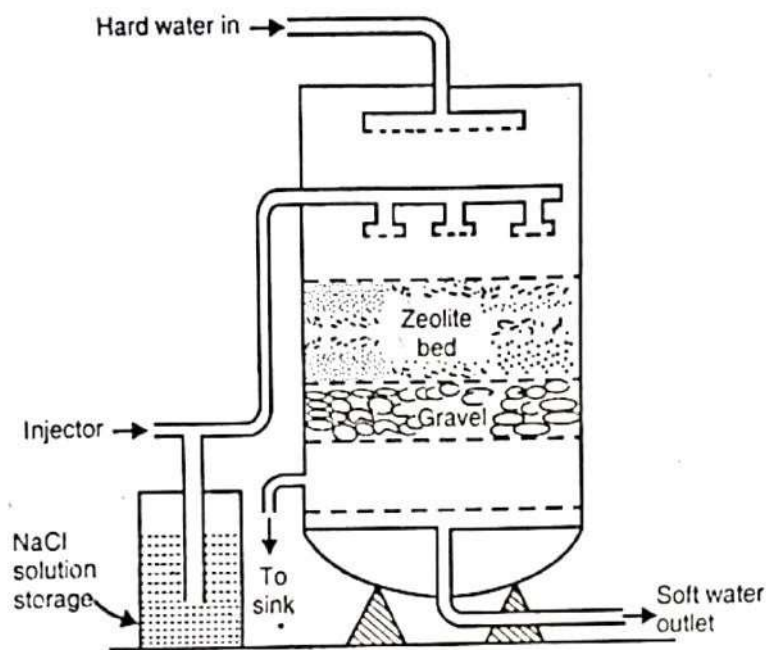
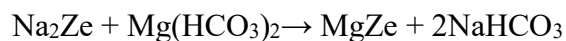
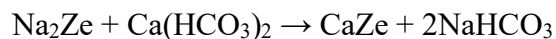
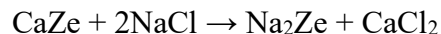


Fig.: Zeolite Softner

Regeneration:

When the zeolite bed becomes exhausted it requires regeneration. This is achieved by passing 10% NaCl solution through it.



Advantages:

1. Almost complete removal of hardness (10ppm)
2. It is compact
3. Requires only less time for softening
4. No sludge formation since no precipitate is formed
5. Can work under pressure also.

Disadvantages:

1. More sodium salt concentration in softened water.
2. Turbidity containing water cannot be used
3. Process exchange only Ca^{2+} and Mg^{2+} ions but cannot exchange HCO_3^- and CO_3^{2-} ions. So cannot be

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used in boilers.

4. If Fe^{2+} and Mn^{2+} are present in large quantities. They form respective zeolites so zeolites cannot be regenerated
5. Water consisting of high alkalinity or acidity cannot be used because zeolite is decomposed

2. LIME-SODA PROCESS

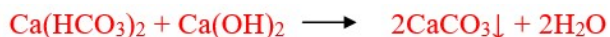
Principle: The lime soda process involves the chemical conversion of all the soluble hardness causing salts by the addition of soda (Na_2CO_3) and lime [$\text{Ca}(\text{OH})_2$] into insoluble precipitates which could easily be removed by settling and filtration.

Functions of lime Removes:

1. Temporary hardness
2. Permanent Mg hardness
3. Dissolved Fe, Al salts
4. Dissolved CO_2 and H_2S gases
5. Free mineral acids Present in water

Lime:

The temporary hardness of water can be removed by hydrated lime. It reacts with $\text{Ca}(\text{HCO}_3)_2$ & $\text{Mg}(\text{CO}_3)_2$ to form insoluble precipitate of calcium carbonate and magnesium hydroxide respectively.



Hydrated lime is also used to remove permanent hardness (magnesium salt impurities only) from water. It reacts with MgSO_4 & MgCl_2 to form insoluble precipitate of magnesium hydroxide.



In above reaction calcium based impurities like CaCl_2 & CaSO_4 also form, which are soluble in water. Therefore calcium based impurities (CaCl_2 & CaSO_4) are not removed by lime treatment.

Soda:

Sodium carbonate (Na_2CO_3) is used to remove permanent hardness of water which caused by MgSO_4 & MgCl_2 or CaCl_2 & CaSO_4 .

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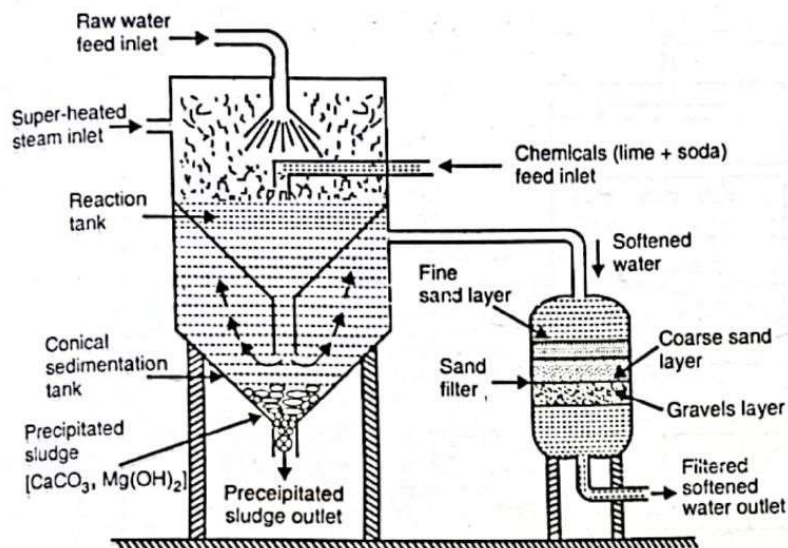
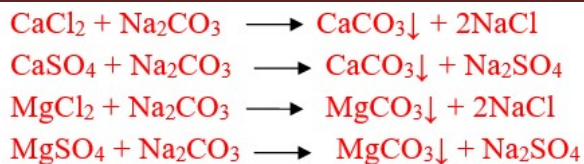


Fig. Continuous hot lime-soda softner

Hot lime soda process:

Calculated quantity of lime and soda are mixed with water at 80 to 150°C.

Advantages of high temperature:

- The reaction proceeds faster.
- Softening capacity is increased.
- No coagulant needed as the precipitate and sludge formed settle down rapidly.
- Much of the dissolved gases are driven out of water.
- Viscosity of soften water is lower, so filtration of water becomes easier.
- This process produces water of comparatively low residual hardness 15 to 30 ppm.

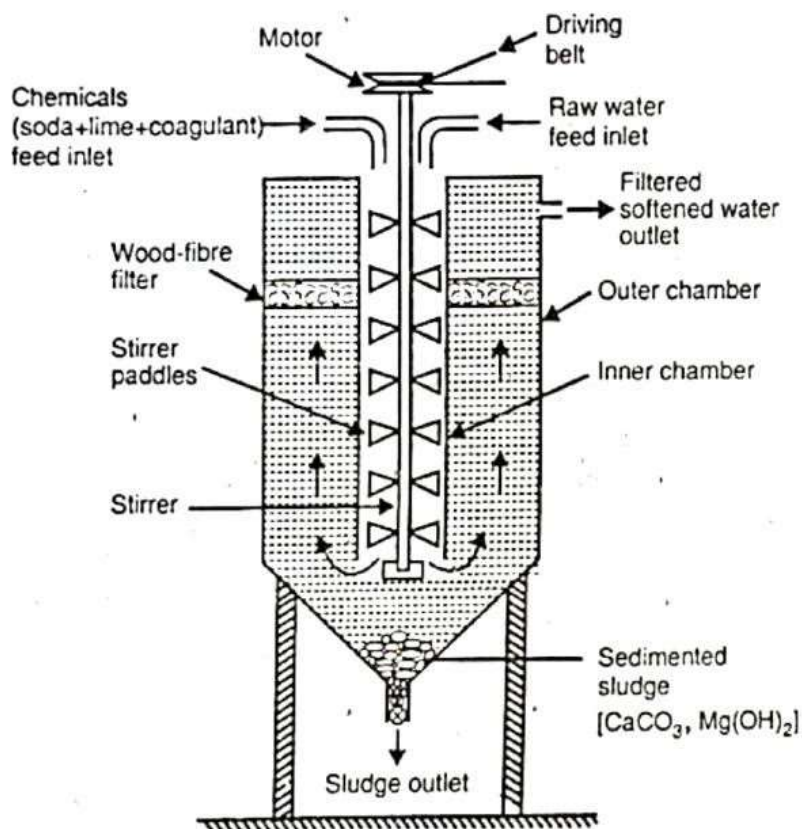
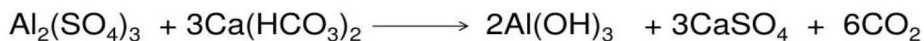


Fig. Continuous cold lime-soda softner

Cold lime soda process:

Calculated quantity of lime and soda are mixed with water at room temperature the precipitates formed are finely divided, so they do not settle down easily. Therefore, It is essential to add small amounts of coagulants (like alum, aluminum sulphate, sodium aluminates *etc*).

Use of sodium aluminate as coagulant also helps the removal of silica as well as oil if present in water..



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Now 100 parts by mass of CaCO_3 are equivalent to : (i) 74 parts of Ca(OH)_2 , and (ii) 106 parts of Na_2CO_3 .

∴ Lime requirement for softening

$$= \frac{74}{100} \left[\begin{array}{l} \text{Temp. Ca}^{2+} + 2 \times \text{Temp. Mg}^{2+} + \text{Perm. (Mg}^{2+} + \text{Fe}^{2+} + \text{Al}^{3+}) \\ + \text{CO}_2 + \text{H}^+ (\text{HCl or H}_2\text{SO}_4) + \text{HCO}_3^- - \text{NaAlO}_2 \\ \text{all in terms of CaCO}_3 \text{ eq} \end{array} \right]$$

∴ Soda requirement for softening

$$= \frac{106}{100} \left[\begin{array}{l} \text{Perm. (Ca}^{2+} + \text{Mg}^{2+} + \text{Al}^{3+} + \text{Fe}^{2+}) + \text{H}^+ (\text{HCl or H}_2\text{SO}_4) - \text{HCO}_3^- \\ \text{all in terms of CaCO}_3 \text{ eq} \end{array} \right]$$

Advantages:

1. Economical
2. Hot lime soda process is much faster than the cold lime soda process
3. During this process pH value of water is increased hence the corrosion of pipe is reduced
4. Besides the removal of hardness, the quantity of minerals in water is also reduced
5. Due to alkaline nature of water, amount of pathogenic bacteria in water are also removed
6. Requires less amount of coagulants

Disadvantages:

1. The softened water is not completely free from hardness (15-30ppm of hardness still remains)
2. Disposal of large amount of sludge is a problem
3. Careful operation and skilled supervision is required for efficient treatment of water

Comparison of Permutit Process with Lime-Soda Process:

<i>Permutit method</i>	<i>Lime-soda method</i>
1. Water of 10-15 ppm residual hardness is obtained.	Water of, generally, 15-50 ppm hardness is obtained.
2. Treated-water contains larger amount of sodium salts than in original raw water.	Treated-water contains lesser amount of sodium salts.
3. Cost of plant and material is higher.	Capital cost is lower.
4. Operation expenses are lower.	Operation expenses are higher.
5. It cannot be used for treating acidic water, because the permutit material undergoes disintegration.	There are no such limitations.
6. The plant occupies less space.	Plant occupies more space.
7. The raw water to be softened must be free from suspended matter ; otherwise the pores of permutit material are blocked and the bed loses its exchange capacity.	There are no such limitations.
8. It can operate under pressure and can be made fully automatic.	It cannot operate under pressure.
9. It involves no problem of settling, co-agulation, filtration and removal of the sludges and precipitates.	It involves difficulty in settling, co-agulation, filtration and removal of precipitates.
10. Control test comprises only in checking the hardness of treated-water.	In order to meet the changing hardness of incoming water, frequent control and adjustment of reagents is needed.
11. Treated-water contains more dissolved solids.	Treated-water contains lesser dissolved solids.

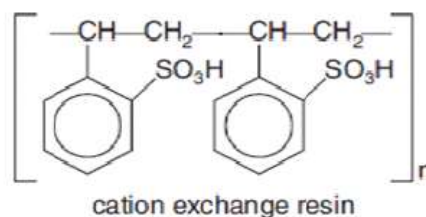
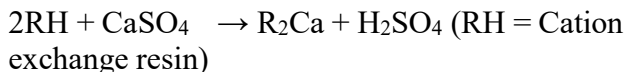
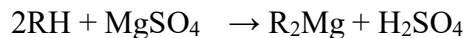
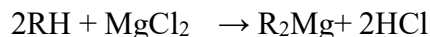
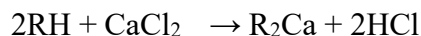
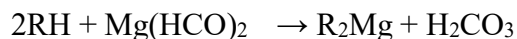
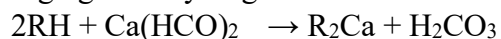
3. ION EXCHANGE / DEMINERALIZATION PROCESS

Ion exchange process is also known as demineralization process. Ion-Exchange resins are insoluble. Cross linked long chain organic polymers with a microporous structure, and the “functional Groups” attached to the chains are responsible for the ion-exchanging properties. Resins with acidic functional group are capable of exchanging H^+ ions with other cations. Resins with basic functional groups are capable of exchanging OH^- ions with other anions.

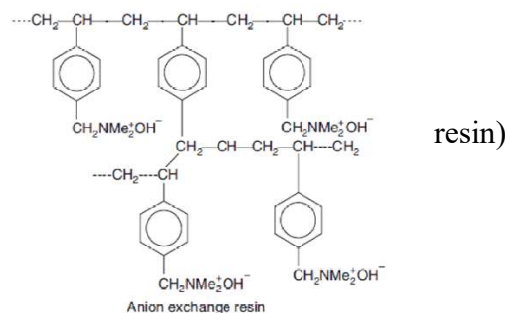
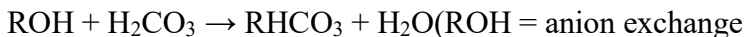
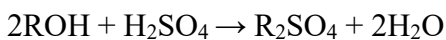
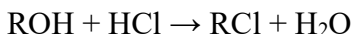
Resins are classified as:

- i. Cation Exchange Resins
- ii. Anion Exchange Resins.

- i. Cation Exchange Resins:** Cation exchange resins are styrene divinyl benzene co-polymers, which on sulphonation (or) carboxylation, which contains $-COOH$, $-SO_3H$ functional groups which responsible for exchanging their hydrogen ions with cations in water.



- ii. Anion Exchange Resins:** Anion exchange resins are Phenol formaldehyde (or) amine formaldehyde copolymers, which contains amino or basic functional groups which responsible for exchanging their OH^- ions with anions in water.



In ion-exchange process, hard water is allowed to pass through cation exchange resins, which remove Ca^{+2} and Mg^{+2} ions and exchange equivalent amount of H^+ ions. Anions exchange resins remove bicarbonates, chlorides and sulphates from water exchange equivalent amount of OH^- ions.

Thus by passing hard water through cation hardness is observed by the following reactions. H^+ and OH^- ions, thus released in water from respective cation and anion exchange columns, get combined to produce water molecules.

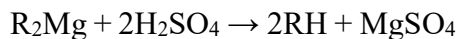
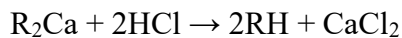


The water coming out from the exchanger is ion free from anions and cations. Thus water of zero hardness is obtained.

Regeneration: When cation exchanger losses capacity of producing H^+ ions and exchanger losses capacity of producing OH^- ions, they are said to be exhausted. The exhausted cation exchanger is

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regenerated by passing it through dilute sulphuric acid.



The exhausted anion exchanger is regenerated by passing a dilute solution of NaOH.

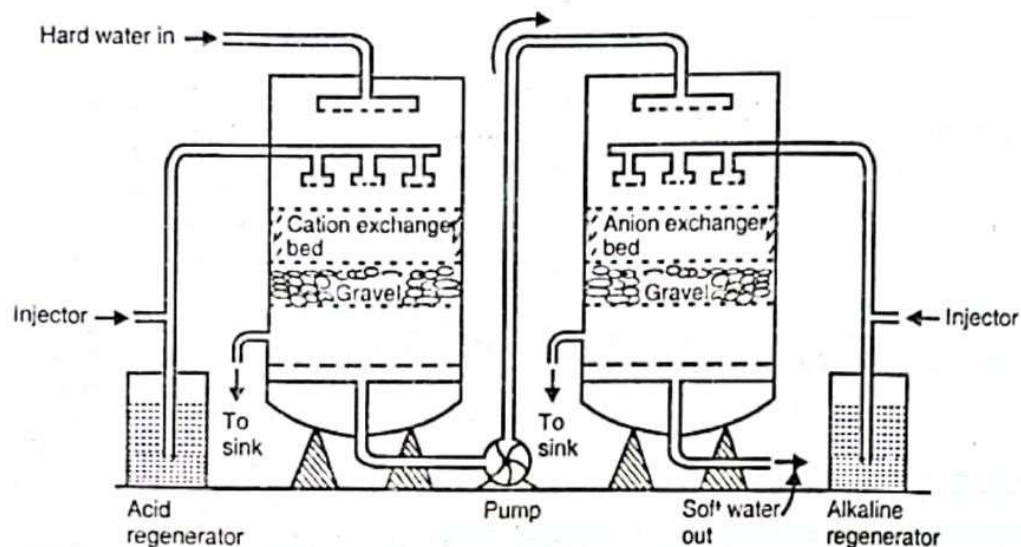
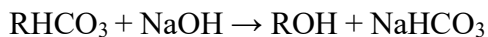
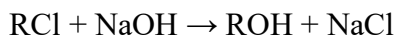
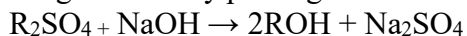


Fig. Ion Exchange Process

Merits of Ion-exchange process:

- The process can be used to soften highly acidic or alkaline water.
- It produces water of very low hardness (2ppm)
- So it is very good for treating water for use in high-pressure boilers.

Demerits of Ion-exchange process:

- The equipment is costly and more expensive chemicals are needed.
- If water contains turbidity, the output of the process is reduced. The turbidity must be below 10ppm; else it has to be removed by coagulation and filtration.

4. DESALINATION OF BRACKISH WATER –REVERSE OSMOSIS

The process of removing common salt (Sodium Chloride) from the water is known as **desalination**.

The water containing dissolved salts with a salty or brackish taste is called **brackish water**.

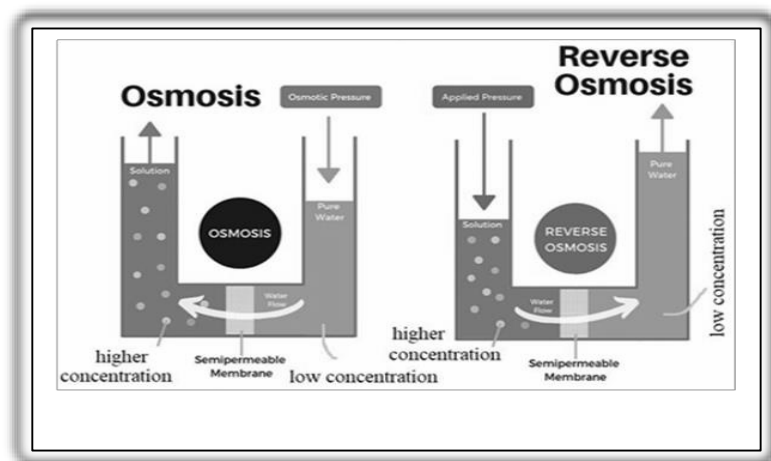
Depending upon the quantity of dissolved solids, water is graded as:

- i. Fresh Water: Contains less than 1000 ppm of dissolved solids.
- ii. Brackish Water: Contains more than 1000 ppm to less than 35000 ppm of dissolved solids.
- iii. Sea Water: Contains more than 35000 ppm of dissolved solids.

Sea water and brackish water can be made available as drinking water through the desalination process. Desalination is carried out either by **reverse osmosis** or **electro dialysis**.

Reverse Osmosis:

When two solutions of unequal concentrations are separated by a semi permeable membrane, flow of solvent takes place from dilute to concentrate sides, due to osmosis. If, however a hydrostatic pressure in excess to osmotic pressure is applied on the concentrated side, the solvent flow is reversed, i.e, solvent is forced to move from concentrated side to dilute side across the membrane. This is the principle of reverse osmosis (RO).



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Thus in reverse osmosis method, pure solvent is separated from its contaminants, rather than removing contaminants from the water. The membrane filtration is sometimes also called super-filtration or hyper filtration.

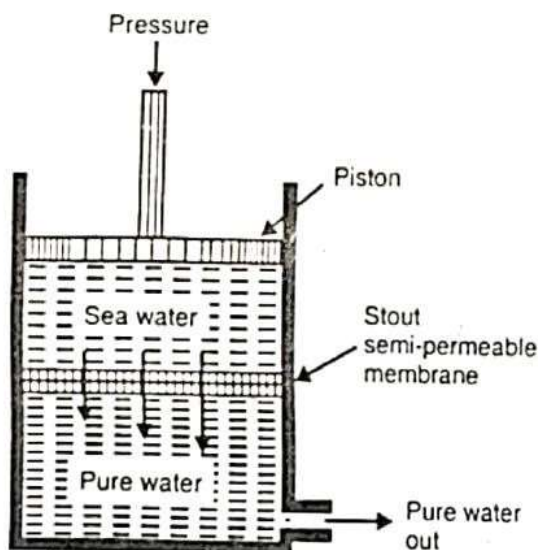


Fig: Reverse Osmosis Cell

Method:

In this process, pressure is applied to the sea water or impure water to force the pure water content of it out the semi-permeable membrane, leaving behind the dissolve solids. The principle of reverse osmosis as applied for treating saline/sea water as illustrated in fig. The membrane consists of very thin film of cellulose acetate, affixed to either side of a perforated tube. However, more recently superior membranes made of polymethacrylate and polyamide polymers have come into use.

Advantages:

1. Reverse osmosis possesses distinct advantages of removing ionic as well as non-ionic, colloidal and high molecular weight organic matter.
2. It removes colloidal silica, which is not removed by demineralization.
3. The maintenance cost is almost entirely on the replacement of the semi permeable membrane.
4. The life time of membrane is quite high, about 2 years,
5. The membrane can be replaced within a few minutes, thereby providing nearly uninterrupted water supply.
6. Due to low capital cost, simplicity, low operating cost and high reliability, the reverse osmosis is gaining grounds at present for converting sea water into drinking water and for obtaining water for very high – pressure boilers.
